

UNITS AND CONSTANTS — CHECK BEFORE ANY NUMBER GOES IN		
QUANTITY	WHAT THE FORMULA NEEDS	THE TRAP
temperature	KELVIN always. K = °C + 273	not 273.15 in this module
gas volume	m³ dm ³ ÷ 1000	1 dm ³ = 1 L = 1000 cm ³ = 0.001 m ³
gas pressure	Pa kPa × 1000	1 atm = 101 325 Pa
solution volume	litres for M = n ÷ V	but mL cancels in M.V. = M.V. and in titration
R	8.314 m ³ Pa K ⁻¹ mol ⁻¹	quoted in the question; the only value used
N_A	6.02 × 10 ²³ per mol	
molar volume	24 dm ³ /mol at 20°C, 1 atm	recall only. Calculations go through PV = nRT
entropy against enthalpy	S° tables in J, ΔH in kJ	divide S by 1000 before combining
electrolysis time	seconds for q = It	convert back to hours only at the end
answers	thermodynamics 2 dp ; elsewhere match the data	state the unit every time

MOLES — EVERYTHING GOES THROUGH THE MIDDLE		
LOOKS LIKE	MOVE	WORKED
mass given	n = m ÷ M	102 ÷ 80.91 = 1.26 mol
moles given, mass wanted	m = n × M	0.80 × 119.00 = 95.20 g
“how many molecules”	N = n × 6.02 × 10 ²³	2.00 mol ÷ 1.20 × 10 ⁻²⁴
molar mass of a formula	count every atom; a number outside a bracket multiplies all of it	Al ₂ (SO ₄) ₃ : the 3 hits S and 4 O = 342.17
unknown element	1. get n from the mole ratio 2. M = mass ÷ n 3. subtract the known atoms	110.98 – 2(35.45) = 40.08 → Ca
before reading any ratio	BALANCE the equation	C ₂ H ₂ + 8O ₂ → 5CO ₂ + 6H ₂ O

GASES		
LOOKS LIKE	MOVE	WORKED
moles change, T and P fixed	Avogadro V ₁ /n ₁ = V ₂ /n ₂	5 mol/50 L → 6 mol = 60 L
P and V, T fixed	Boyle P ₁ V ₁ = P ₂ V ₂	3 atm 330 mL → 1 atm 990 mL
V and T, P fixed	Charles V ₁ /T ₁ = V ₂ /T ₂ , T in K	25→70°C = 298→343 K, up 15.10%
anything with n, or a mixture	PV = nRT	n = PV/RT V = nRT/P
P AND T both change	no combined gas law in A105. Take the ratio	V ₂ /V ₁ = (P ₁ /P ₂) (T ₂ /T ₁)
identify a gas from mass + PVT	n = PV/RT, then M = mass ÷ n	0.71/0.0100 = 71 → Cl ₂

SOLUTIONS, DILUTION AND TITRATION		
LOOKS LIKE	MOVE	WORKED
mol/L, molarity, [X]	M = n ÷ V, V in litres	0.100 ÷ 0.250 = 0.400 M
the other three units	% mass ÷ whole solution g/L ÷ volume mol/kg ÷ SOLVENT in kg	only molality divides by the solvent
“diluted to”, a volumetric flask	M ₁ V ₁ = M ₂ V ₂	V₂ is the FLASK size , not the water added
serial dilution	factors multiply	10 × 10 × 10 = 1000

SOLUTIONS, DILUTION AND TITRATION		
LOOKS LIKE	MOVE	WORKED
a burette table	1. drop every Rough trial 2. final – initial, per trial 3. average the accurate ones 4. n = MV on the KNOWN side 5. apply the mole ratio, divide by the other volume	the bigger coefficient takes the bigger mole count 2 NaOH : 1 H ₂ SO ₄ → ×2
pH given instead of a concentration	[H ⁺] = 10 ^{-pH}	monoprotic strong acid + that IS its molarity
why NaOH must be standardised	it is hygroscopic and absorbs CO₂	its mass is unreliable, so titrate against a known acid

LIMITING REAGENT		
LOOKS LIKE	MOVE	WORKED
two amounts given	1. for each reactant, the product it alone gives 2. the smaller one limits	4 CO (1:1)→4, 6 H ₂ (2:1)→3 → H ₂
the classic error	the smaller STARTING amount is not always limiting	Zn 0.0642 vs CuSO ₄ 0.0500 at 1:1 → CuSO ₄
“how much remains”	start – consumed	0.60 – 0.40 = 0.20 mol

ENERGY 1 — HESS AND ENTHALPY		
LOOKS LIKE	MOVE	WORKED
“given that” plus two or three equations	1. write the TARGET equation 2. species already correct? keep 3. wrong side? reverse , flip the sign of ΔH 4. coefficient wrong? scale , multiply ΔH 5. add, cancel, sum the ΔH	reverse and scale are independent +966 reversed then halved = +483
a table of Δ_fH	Σn(products) – Σn(reactants)	an ELEMENT is zero , including H ₂ , N ₂ , O ₂ , Cl ₂ , Br ₂ , I ₂
a table of bond enthalpies	ΔH = broken (+) + formed (–)	reactants first – the opposite way round
bond counting	each CO ₂ holds 2 C=O, each H ₂ O 2 O–H	3 CO ₂ = 6 bonds, not 3
“gives out 81.67 kJ”	ΔH = –81.67	the minus sign is never written in the question
a combustion that came out positive	the two terms are the wrong way round	combustion is always exothermic

ENERGY 2 — ENTROPY AND FREE ENERGY		
LOOKS LIKE	MOVE	WORKED
“increase or decrease in entropy”	1. count particles : more on the right → up 2. compare states : s < l < aq < g	both tests – one question has equal particles and turns on state alone
a table of S°	ΣS(prod) – ΣS(react)	elements are NOT zero here. O ₂ = 205.0
ΔH and ΔS given, “spontaneous”	1. T to kelvin 2. S in J? divide by 1000 3. ΔG = ΔH – TΔS	<0 spontaneous, >0 not, =0 equilibrium
the double negative	–T × (–ΔS) is plus	–92.2 + 59.30 = –32.90
“at what temperature”	Set ΔG = 0, SO T = ΔH ÷ ΔS	6.01 ÷ 0.0220 = 273.2 K

SPONTANEITY — THE FOUR CASES		
	ΔH NEGATIVE	ΔH POSITIVE
ΔS positive	spontaneous at ALL T	spontaneous at HIGH T
ΔS negative	spontaneous at LOW T	NEVER spontaneous

SPONTANEITY — THE FOUR CASES		
	ΔH NEGATIVE	ΔH POSITIVE
why	only TΔS changes with temperature	same signs → temperature cannot flip the verdict

ATOMS, IONS AND ISOTOPES		
LOOKS LIKE	MOVE	WORKED
mass number A, atomic number Z	protons = Z neutrons = A – Z	U-238: 92 p, 146 n
a neutral atom	electrons = protons	
an ion	positive → lost electrons negative → gained electrons	Ca ²⁺ has 20 p, 18 e
isotopes	same Z, different A	same chemistry, different mass
relative atomic mass from abundances	A _r = Σ(mass × %) ÷ 100	a weighted average, so it is rarely a whole number
electron configuration	fill 1s 2s 2p 3s 3p 4s 3d 4p	4s fills BEFORE 3d, and empties first too
valence electrons	the outermost shell only	Group number gives it for the main groups

BONDING AND POLARITY		
LOOKS LIKE	MOVE	WORKED
metal + non-metal	ionic : electrons transferred	giant lattice, high melting point, conducts when molten or aqueous
non-metal + non-metal	covalent : electrons shared	
Δ(electronegativity)	< 0.5 non-polar covalent 0.5 to 2.0 polar covalent > 2.0 ionic	3.98 – 2.96 = 1.02 → polar covalent
is the MOLECULE polar	polar bonds are not enough — the SHAPE must be asymmetric	CO ₂ has polar bonds and is non-polar; the dipoles cancel
strongest intermolecular force	H bonded to N, O or F → hydrogen bonding else polar → dipole-dipole else London	more hydrogen bonds → more energy to separate
periodic trends	across : electronegativity up, radius down down : electronegativity down, radius up	F is the most electronegative element

VSEPR — SHAPE FROM THE ELECTRON PAIRS			
DOMAINS PAIRS	LONE	SHAPE	ANGLE
2 0		linear	180°
3 0		trigonal planar	120°
3 1		bent	< 120°
4 0		tetrahedral	109.5°
4 1		trigonal pyramid	107.5°
4 2		bent	104.5°
5 0		trigonal bipyramid	120° and 90°
6 0		octahedral	90°
6 1		square pyramid	< 90°
6 2		square planar	90°
getting there	1. valence electrons on the central atom 2. subtract one per bonded atom 3. what is left ÷ 2 = lone pairs 4. domains = bonds + lone pairs		BrF ₅ : 7 – 5 = 2 → 1 lone pair, 6 domains → square pyramid
why an angle shrinks	lone pair repulsion is stronger than bond pair		each extra lone pair squeezes the angle further

“FIND THE PH OF...” — IDENTIFY THE CASE FIRST		
WHAT IS PRINTED	CASE	ROUTE
single arrow →, a named acid, no K	strong acid	$[H^+] = n \times [acid]$, then $pH = -\log[H^+]$
NaOH , KOH, “alkaline”	strong base	two steps. $pOH = -\log[OH^-]$ then $pH = 14 - pOH$
double arrow ⇌, or K_a quoted	weak acid	the % dissociated is always supplied
a % dissociated figure	weak acid, direct	$[H^+] = \% \times [acid]$, then $-\log$
two solutes, second is the Na salt of the first	BUFFER	$pH = pK_a + \log([salt] \div [acid])$
weak base plus its salt	alkaline buffer	gives pOH. Then $14 - pOH$
strong acid with its salt	not a buffer	treat as the strong acid
weak acid with an unrelated salt	not a buffer	treat as the weak acid
the two concentrations are equal	$pH = pK_a$ exactly	$\log(1) = 0$

PH ARITHMETIC		
LOOKS LIKE	MOVE	WORKED
the four relations	$pH = -\log[H^+]$ $[H^+] = 10^{-pH}$ $pOH = -\log[OH^-]$ $[OH^-] = 10^{-pOH}$ $pH + pOH = 14$	any quantity reaches any other in two steps
monoprotic strong: HCl HNO ₃ HBr	$[H^+] = 1 \times [acid]$	$0.03\text{ M} \rightarrow pH\ 1.52$
diprotic strong: H ₂ SO ₄	$[H^+] = 2 \times [acid]$	$0.015\text{ M} \rightarrow 0.030 \rightarrow pH\ 1.52$
H₂CO₃, H₂S	diprotic but WEAK	the doubling is for strong diprotic only
weak acid, % given	$[H^+] = \% \times c$, then $K_a = [H^+][A^-] \div [HA]$	0.010 M at $5\% \rightarrow 5.0 \times 10^{-4} \rightarrow pH\ 3.30$
a weak and a strong acid at the same pH	pH cannot tell them apart. Only K_a can	0.010 weak and 0.0005 strong both give 3.30

BUFFERS		
LOOKS LIKE	MOVE	WORKED
is it a buffer	1. is the acid or base weak ? 2. are both partners present? 3. is the salt the conjugate of that same acid?	all three must hold
acidic buffer	$pH = pK_a + \log([salt] \div [acid])$	salt on top. $pK_a = -\log K_a$
alkaline buffer	$pOH = pK_b + \log([salt] \div [base])$	gives pOH, then $14 - pOH$
direction check	more salt than acid → pH above pK_a	the other way means the ratio is inverted
add a little acid	the basic partner mops it up	$H^+ + CH_3COO^- \rightarrow CH_3COOH$
add a little base	the acidic partner mops it up	$CH_3COOH + OH^- \rightarrow H_2O + CH_3COO^-$
“buffering capacity”	higher concentration of both, as near equal as possible	two buffers can share a pH and differ in capacity
pick an acid for a target pH	at equal concentrations $pH = pK_a$	target $8.6 \rightarrow$ the acid with $pK_a\ 8.60$
a required ratio	$[salt] \div [acid] = 10^{(pH - pK_a)}$	blood $10^{1.3} \rightarrow$ about $20:1$

EQUILIBRIUM		
LOOKS LIKE	MOVE	WORKED
“write the K_c expression”	1. balance first 2. products over reactants 3. each raised to its coefficient 4. multiply, never add	$CH_4 + 2H_2S \rightleftharpoons CS_2 + 4H_2$ $K_c = [CS_2][H_2]^4 \div ([CH_4][H_2S]^2)$
“value and units”	M^n , Δn = powers above – below	$\Delta n = 0 \rightarrow$ no units
Q_c against K_c	same algebra	Q at any instant; equal to K only at equilibrium
“essentially only product”	$K > 10^2$ product, $< 10^{-2}$ reactant	the boundary is 100. $K = 97$ is comparable
add reactant / remove product	shifts right	K unchanged
raise pressure	shifts to fewer moles of gas	equal moles → no shift. K unchanged
raise temperature	shifts the way that absorbs heat	the ONLY change that alters K
catalyst	no shift at all	no change to position, K or yield
heat written in the equation	treat heat as a species	products + heat = exothermic, so heating shifts left

KINETICS		
LOOKS LIKE	MOVE	WORKED
“why does X speed it up”	every reason ends the same way: more successful collisions	surface area, concentration, pressure → more frequent collisions
temperature	the only factor that does both	more collisions and a bigger proportion above E_a
catalyst	an alternative route of lower E_a	not used up; k increases
a Maxwell-Boltzmann figure	able to react = area right of E_a	temperature SHIFTS the curve; concentration only RAISES it; a catalyst moves the E_a LINE
rate = $k[A]^m[B]^n$	orders come from experiment , never the coefficients	the one exception: an ELEMENTARY step
order from a rates table	find two runs where only one concentration changed	rate same → 0 doubles → 1 quadruples → 2
order from half-life	$t_{1/2}$ constant → first halving → zero doubling → second	constant 12 s each time → first order
units of k	$M^{(1 - \text{overall order})} s^{-1}$	overall $1 \rightarrow s^{-1}$ overall $3 \rightarrow M^{-2} s^{-1}$
two temperatures	$\ln(k_2/k_1) = E_a/RT_1 - E_a/RT_2$	°C → K first. R in kJ if E_a in kJ

OXIDATION STATES — APPLY IN THIS ORDER		
LOOKS LIKE	MOVE	WORKED
uncombined element	0 , and stop	S_8 , O_2 , P_4 , $Na \rightarrow$ all 0
whole species is one atom with a charge	= the charge, and stop	$Fe^{3+} \rightarrow +3$
Group 1, 2, F	+1, +2, –1 , never bend	$Mg(OH)_2 \rightarrow Mg = +2$ immediately
H	+1 , unless bonded to a metal → –1	NaH , CaH_2
O	–2 , unless peroxide → –1	H_2O_2 and $O_2^{2-} \rightarrow 0 = -1$
Cl, Br, I	–1 in a binary compound. If O or F is also there, do NOT assign	$HClO_4 \rightarrow Cl$ is the unknown
whatever is left	$\Sigma(\text{count} \times \text{state}) =$ total charge	neutral → 0 polyatomic ion → its charge
a fast self-check	a state above +7 means a charge was forgotten	MnO_4^- summed to 0 gives +8, which is impossible

REDOX AND CELLS		
LOOKS LIKE	MOVE	WORKED
“what is oxidised / reduced”	1. assign states on both sides 2. find the atoms whose number changed 3. up = oxidised, down = reduced	nothing changed → not a redox reaction
“the oxidising agent”	the thing that gets REDUCED	the two words are always opposites
balance in acid	1. split into two half-equations 2. balance all but O and H 3. add H₂O for O, then H⁺ for H 4. add electrons to balance charge 5. scale so the electrons cancel, then add	check: same atoms and same total charge each side
balance in base	do all of the above via H⁺ first , then add the same number of OH^- to both sides	starting from OH^- is much harder
galvanic or electrolytic	no battery, spontaneous, E° positive → galvanic power source, driven, E° negative → electrolytic	
what NEVER changes	anode = oxidation, cathode = reduction, in BOTH	only the signs swap
the signs	galvanic: anode –, cathode +	electrolytic: anode +, cathode –
which one is reduced	the more positive E°	-0.28 beats -0.74
“calculate E°_{cell} ”	$E^\circ_{\text{cell}} = E^\circ_{\text{ox}} + E^\circ_{\text{red}}$	reverse the oxidised half and flip its sign
a scaled half-equation	E° is NOT multiplied. Volts are per electron	scale the equation, leave +0.80 alone
electrons flow	always anode to cathode	through the external wire

ELECTROLYSIS — THE CHAIN		
LOOKS LIKE	MOVE	WORKED
mass from current and time	1. $Q = I \times t$, t in seconds 2. $n(e^-) = Q \div 96500$ 3. divide by the electrons in the half-equation 4. $\times M$ for grams	2 A for $1\text{ h} \rightarrow Q = 7200 \rightarrow 0.0746\text{ mol } e^-$ $Na^+ + e^-$ is $1:1 \rightarrow 1.72\text{ g}$
“how long” or “what current”	run the chain backwards	$\text{mol} \times 96500 \div Q \rightarrow s \div 3600$
plating or refining	the object is the CATHODE	metal always deposits at the cathode
if the question prints an E° value	use the question, not the table	the tables disagree on Ni (-0.25 vs -0.28)

NOT EXAMINED, AND KNOWN MISPRINTS	
Outside A105: STP and 22.4 L/mol; Gay-Lussac; the combined gas law; percent composition; empirical and molecular formula; ppm; ICE tables; K_p ; K_w as a symbol; the “x is small” approximation; pH of a salt solution; calorimetry ($q = mc\Delta T$); cell notation such as $Zn Zn^{2+} Cu^{2+} Cu$. Misprints in the course material: one equilibrium reads $PCl_5 \rightleftharpoons PCl_5 + Cl_2$ (the left should be PCl_3); one reads $SO_2 + O_2 \rightleftharpoons 2SO_3$ unbalanced (should be $2SO_2$); the activity series prints Mg twice where the second should be Mn ; a car-battery equation prints $PbSO_4$ for PbSO₄ ; one iron potential is shown as 0.44 V when it is –0.44 V . pK_a of ethanoic acid appears as 4.76 in one place and 4.74 elsewhere. Use 4.74 . A periodic table is NOT supplied for every question — one electrolysis question needs the molar masses of Hg and Cl and gives neither.	